

PREFACE FOR FACULTY

This lecture establishes how discrete LTI systems modify the magnitude and phase of input signals. A major conceptual pillar in DSP is the **Eigenfunction Property**: When a complex exponential $e^{j\omega n}$ passes through an LTI system, the output is the same complex exponential scaled by the complex frequency response $H(e^{j\omega})$.

Pedagogical Strategy:

1. Derive the eigenfunction property: $\mathcal{T}e^{j\omega n} = H(e^{j\omega})e^{j\omega n}$.
 2. Decompose $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}$.
 3. Differentiate between **Phase Delay** $\tau_p(\omega) = -\frac{\angle H(e^{j\omega})}{\omega}$ and **Group Delay** $\tau_g(\omega) = -\frac{d}{d\omega}\angle H(e^{j\omega})$.
 4. Explain the physical consequence of non-constant group delay: **Dispersion (phase distortion)**, which spreads wave packets in time.
 5. Define the condition for strictly linear phase: $\angle H(e^{j\omega}) = -\alpha\omega \implies \tau_g(\omega) = \alpha = \text{constant}$.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the steady-state response of an LTI system to sinusoidal and multi-tone inputs.
 2. **Calculate** the magnitude response $|H(e^{j\omega})|$, phase response $\theta(\omega)$, phase delay $\tau_p(\omega)$, and group delay $\tau_g(\omega)$.
 3. **Analyze** the distortion introduced by non-linear phase filters on broadband and modulated communication signals.
 4. **Determine** the conditions under which a digital filter provides generalized linear phase.
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2. MATHEMATICAL FOUNDATIONS

2.1 The Eigenfunction Property

Let input $x[n] = e^{j\omega n}$ for all $-\infty < n < \infty$. The system output is:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = \left(\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \right) e^{j\omega n} = H(e^{j\omega})e^{j\omega n}$$

Where $H(e^{j\omega}) = \mathcal{F}h[n]$ is the **Frequency Response** of the system.

For a real sinusoidal input $x[n] = A \cos(\omega_0 n + \phi)$:

$$y[n] = A|H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0}))$$

2.2 Phase Delay vs. Group Delay

- **Phase Delay** $\tau_p(\omega)$: The time delay experienced by individual sinusoidal carrier frequencies:

$$\tau_p(\omega) = -\frac{\angle H(e^{j\omega})}{\omega} = -\frac{\theta(\omega)}{\omega}$$

- **Group Delay** $\tau_g(\omega)$: The time delay experienced by the envelope/information of a narrowband wave packet (derivative of phase):

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} = -\frac{d}{d\omega} [\angle H(e^{j\omega})]$$

2.3 Condition for Distortionless Transmission

For a system to transmit a signal without wave shape distortion:

$$y[n] = Ax[n - n_d]$$

In the frequency domain:

$$Y(e^{j\omega}) = Ae^{-j\omega n_d} X(e^{j\omega}) \implies H(e^{j\omega}) = Ae^{-j\omega n_d}$$

1. **Magnitude Response:** $|H(e^{j\omega})| = A = \text{constant}$ (No amplitude distortion).
2. **Phase Response:** $\theta(\omega) = -\omega n_d$ (Strictly linear phase).
3. **Group Delay:** $\tau_g(\omega) = n_d = \text{constant}$ (All frequencies delayed equally).

3. WORKED NUMERICAL EXAMPLES

Example 4.1: Frequency Response & Delay of a Moving-Average Filter

Problem: A 3-point moving average filter has impulse response $h[n] = \frac{1}{3} \underset{\uparrow}{1}, 1, 1$. (a) Find $H(e^{j\omega})$. (b) Determine the magnitude response $|H(e^{j\omega})|$ and phase response $\theta(\omega)$. (c) Compute the phase delay $\tau_p(\omega)$ and group delay $\tau_g(\omega)$.

Solution: (a)

$$H(e^{j\omega}) = \frac{1}{3}(1 + e^{-j\omega} + e^{-j2\omega}) = \frac{1}{3}e^{-j\omega}(e^{j\omega} + 1 + e^{-j\omega}) = \frac{1}{3}e^{-j\omega}(1 + 2\cos\omega)$$

(b)

- Amplitude function: $A(\omega) = \frac{1+2\cos\omega}{3}$.
- Magnitude: $|H(e^{j\omega})| = \frac{|1+2\cos\omega|}{3}$.
- Phase: $\theta(\omega) = -\omega$ (for $1 + 2\cos\omega > 0$).

(c)

- Phase Delay: $\tau_p(\omega) = -\frac{\theta(\omega)}{\omega} = -\frac{-\omega}{\omega} = 1$ sample.
- Group Delay: $\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} = -\frac{d(-\omega)}{d\omega} = 1$ sample. Since $\tau_g(\omega) = 1$ is constant for all frequencies where $A(\omega) > 0$, the filter has **strictly linear phase**.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Define Phase Delay and Group Delay. Explain why constant group delay is critical in digital communications and biomedical instrumentation. (6 Marks) (b) An LTI system has frequency response:

$$H(e^{j\omega}) = \frac{1 - 0.5e^{-j\omega}}{1 - 0.8e^{-j\omega}}$$

- Find the steady-state output $y[n]$ for $x[n] = 2 + 4 \cos\left(\frac{\pi}{2}n + \frac{\pi}{6}\right)$. (6 Marks)
- Find the group delay $\tau_g(\omega)$ at $\omega = 0$ and $\omega = \pi$. (3 Marks)

Model Answer & Step-by-Step Marking Rubric:

- Part (a):**
 - Mathematical definitions of τ_p and τ_g (2 Marks)
 - Explanation: Variable group delay causes dispersion/pulse broadening leading to Intersymbol Interference (ISI) in comms, and waveform distortion in ECG/EEG (4 Marks)

- Part (b.1):**
 - At DC ($\omega = 0$): $H(e^{j0}) = \frac{1-0.5}{1-0.8} = \frac{0.5}{0.2} = 2.5$. DC output: $y_{DC}[n] = 2 \times 2.5 = 5$. (2 Marks)
 - At $\omega = \pi/2$: $e^{-j\pi/2} = -j$.

$$H(e^{j\pi/2}) = \frac{1 + 0.5j}{1 + 0.8j} = \frac{\sqrt{1 + 0.25}e^{j\arctan(0.5)}}{\sqrt{1 + 0.64}e^{j\arctan(0.8)}} = \frac{1.1180e^{j0.4636}}{1.2806e^{j0.6747}} = 0.8730e^{-j0.2111} \text{ rad}$$

(2 Marks)

- Total output:

$$y[n] = 5 + 4(0.8730) \cos\left(\frac{\pi}{2}n + \frac{\pi}{6} - 0.2111\right) = 5 + 3.492 \cos\left(\frac{\pi}{2}n + 0.3125\right)$$

(2 Marks)

- Part (b.2):**
 - Group delay formula for single-pole single-zero sections: $\tau_g(\omega) = \frac{0.8(0.8 - \cos \omega)}{1 - 1.6 \cos \omega + 0.64} - \frac{0.5(0.5 - \cos \omega)}{1 - \cos \omega + 0.25}$
 - At $\omega = 0$: $\tau_g(0) = \frac{0.8(-0.2)}{0.04} - \frac{0.5(-0.5)}{0.25} = -4.0 - (-1.0) = -3.0 \rightarrow \text{delay} + 3.0$. (1.5 Marks)
 - At $\omega = \pi$: $\tau_g(\pi) = \frac{0.8(1.8)}{3.24} - \frac{0.5(1.5)}{2.25} = 0.444 - 0.333 = 0.111$ samples. (1.5 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Verify frequency response at w = pi/2
w = np.pi / 2
H = (1 - 0.5 * np.exp(-1j * w)) / (1 - 0.8 * np.exp(-1j * w))
print(f"|H(e^{j(pi/2)})| = {np.abs(H):.4f}, Angle = {np.angle(H):.4f} rad")
```